FIITJEE

ALL INDIA INTEGRATED TEST SERIES

PART TEST – III

JEE (Advanced)-2021

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B.**

1. ***Section-A (01– 7, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*

   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section-A (08 – 13, 26 – 31, 44 – 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S). Each multiple choice question with 4 options has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section–B (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answers as numerical value from 0 to 9 and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Three concentric spherical shells have radii r_1, 2r and 3r with charge q_1, q_2 and q_3 respectively. Innermost and outermost shells are earthed. Then

(A) $q_1 + q_3 = -q_2$ (B) $q_1 = -\frac{q_2}{4}$

(C) $\frac{q_3}{q_1} = 3$ (D) $\frac{q_3}{q_2} = -\frac{1}{3}$

2. A particle having mass m and charge q moves along a line under the action of an electric field $E = \alpha - \beta x$ where α and β are positive constants and x is the distance from a point where the particle was initially at rest. Therefore an observer moving with an acceleration $\frac{q\alpha}{m}$.

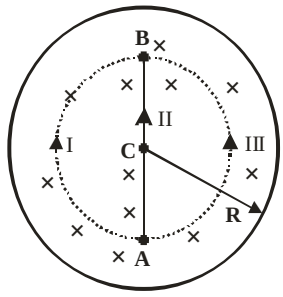
(A) the motion of the particle is oscillatory

(B) the amplitude of the particle is $\frac{\alpha}{\beta}$

(C) the mean position of the particle is at $x = \frac{\alpha}{\beta}$

(D) the maximum acceleration of the particles is $\frac{q\alpha}{m}$

3. Magnetic field is present in a circular region of radius R. Assume the field depends of time given by $B = \alpha_0 t$ [α_0 is positive constant] charge q_0 is moved this region from A to B through different path as shown in figure. Then [(w_I, w_{II}, w_{III}) is work done by induced electric field in path I, II, III.]

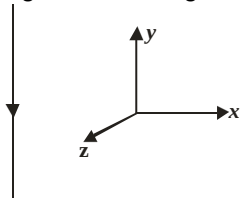


(A) $\frac{\omega_I}{\omega_{III}} = -1$ (B) $\frac{\omega_{II}}{\omega_I} = 0$

(C) ω_{II} is positive (D) ω_{III} is positive

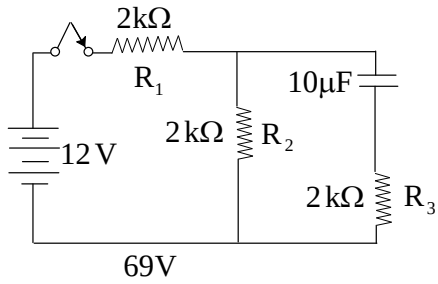
Space for Rough work

4. A infinite length current carrying wire is placed in *x, y* plane parallel to y-axis as shown in figure (all charges are projected in region I from origin. The wire is passing through the points (-4, 0, 0))



(A) A positively charged particle projected along y-axis will get deflected along *x*-axis
(B) If charged particle is projected along positive z-direction from origin then initial acceleration is zero
(C) An electron projected along positive *x*-direction will get deflected along positive y- direction
(D) An electron projected along positive *x* axis will move is circular path.

5. In the circuit shown, the switch is turned on at t = 0, then
(A) at t = 0, current supplied by battery is 4 mA
(B) at t = 0, current in R_3 is 2 mA
(C) in the steady state current supplied by battery is 3 mA
(D) in the steady state current in R_3 is zero



6. A wire of length *l* carrying a current i is bent first in form of an equilateral triangle, the magnetic field at the center of triangle is B_1. The same wire is now bent in the form of a circle; the magnetic field at the centre is now B_2. Each of the coil is kept in a constant magnetic field with its plane perpendicular to magnetic field then τ_1 and τ_2 are torques acting on it respectively

(A) $B_1 > B_2$ (B) $B_2 > B_1$

(C) $\tau_1 = \tau_2 = 0$ (D) $\tau_2 = \tau_1 \neq 0$

7. When the terminals of a cell of e.m.f. 1.5 V are connected to an ammeter of resistance 4Ω. The ammeter reads 0.30 amps. Which of the following statements is/are correct?
(A) The cell is non-ideal
(B) If a 4Ω resistor is also connected across terminals of the cell, the ammeter will read 0.50 amp
(C) If a 4Ω resistor is also connected across the terminals of the cell, one-third of the electrical power generated will dissipate as heat within the cell
(D) If a voltmeter of resistance 4Ω is used to measure potential difference between terminals of the cell, it will read 1.2 V

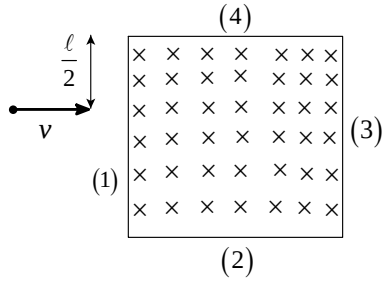
Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions Q. 8, Q. 9 and Q. 10 by appropriately matching the information given in the three columns of the following table.

Magnetic field of confined to a square region of side 'ℓ'
B = magnetic field
m = mass of the particle
q = charge of the particle
v = velocity of the particle
Column (1): Represents charge of the particle
Column (2): Represents velocity of the particle
Column (3): Shows the edge from which the particle comes out



Column – 1		Column – 2		Column – 3	
(I)	$q = q_0$	(i)	$v < \frac{q_0 B_0 \ell}{4m_0}$	(P)	2
(II)	$q = -q_0$	(ii)	$v > \frac{5q_0 B_0 \ell}{4m_0}$	(Q)	1
(III)	$q = 3q_0$	(iii)	$\frac{q_0 B_0 \ell}{4m_0} < v < \frac{5q_0 B_0 \ell}{4m_0}$	(R)	3
(IV)	$q = -\frac{q_0}{2}$	(iv)	$\frac{q_0 B_0 \ell}{2m_0} < v < \frac{5q_0 B_0 \ell}{2m_0}$	(S)	4

8. If $q = q_0$, $B = B_0$, $m = m_0$. Choose the correct option.
(A) (II) (ii) (S) (B) (I) (i) (Q)
(C) (I), (iii) (P) (D) (I) (i) (R)

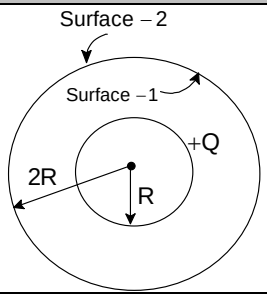
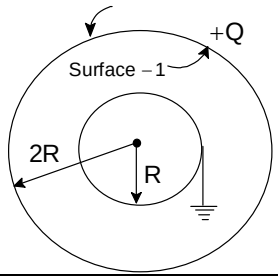
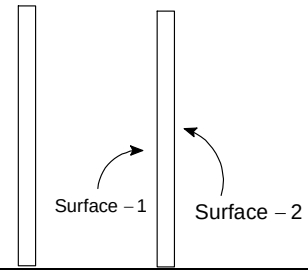
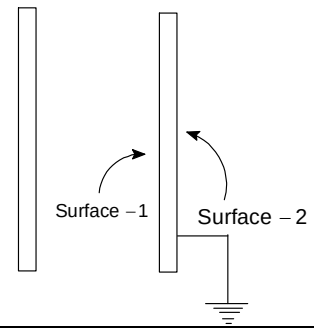
9. If $B = \frac{B_0}{3}$, $m = m_0$ and particle comes out from edge 3, choose the correct option.
(A) (I) (i) (Q) (B) (III) (ii) (R)
(C) (III), (iii) (R) (D) (III) (i) (R)

10. If $B = B_0$ and $m = \frac{m_0}{2}$ and particle comes out from edge 2, choose the correct option.
(A) (II) (i) (P) (B) (III) (i) (P)
(C) (IV), (iii) (P) (D) (IV) (iv) (P)

Space for Rough work

Answer questions Q. 11, Q. 12 and Q. 13 by appropriately matching the information given in the three columns of the following table.

Figure shows total charge given to a conductor.
Here. Column 1 represents arrangement of conductors.
Column 2 represents charge on surface – 1
Column 3 represents charge on surface - 2

Column – 1		**Column – 2**		**Column – 3**	
(I)		(i)	$+Q$	(P)	$+Q$
(II)		(ii)	$-Q$	(Q)	$-Q$
(III)		(iii)	$+\frac{Q}{2}$	(R)	$+\frac{Q}{2}$
(IV)		(iv)	$-\frac{Q}{2}$	(S)	$-\frac{Q}{2}$

11. For part (I) the correct combination is
(A) (I) (i) (P) (B) (I) (ii) (P)
(C) (I), (iii) (R) (D) (I) (iii) (S)

Space for Rough work

12. For part (II) the correct combination is
(A) (II) (i) (Q) (B) (II) (iii) (R)
(C) (II), (iii) (S) (D) (II) (i) (R)

13. For part (III) and (IV) the correct combination is
(A) (IV) (iv) (S) (B) (IV) (iii) (S)
(C) (III), (iv) (R) (D) (III) (iv) (S)

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

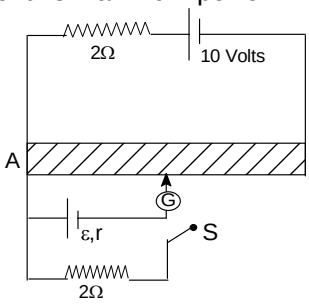
14. Two uniformly charged large plane sheets S_1 and S_2 having charge densities σ_1 and σ_2 $(\sigma_1 > \sigma_2)$ are placed at a distance d parallel to each other. A charge q_o is moved along a line of length a (a < d) at an angle 45° with the normal to S_1 (between sheets). Calculate the work done (in Joule) by the electric field.
(take $a(\sigma_1 - \sigma_2) = 4\sqrt{2}$) and $q_0 = \varepsilon_0$)

15. A long, straight wire carries a current i, A particle having a positive charge q and mass m, kept at a distance x_0 = 2e m, where e is the exponent, from the wire is projected towards it with a speed v. Find the minimum separation between the wire and the particle. $(2\pi mv = \mu_0 qi)$.

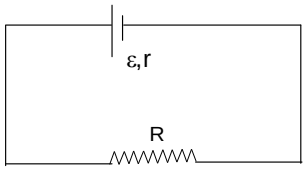
16. A wire of resistance 0.1 ohm/cm is bent to form a square ABCD of side 10 cm. A similar wire is connected between the corners B and D to form the diagonal BD. If 2 volt battery of negligible internal resistance is connected across A and C, calculate the total power dissipated.

Space for Rough work

17. Circuit – 1 shows a potentiometer setup with length of potentiometer being 100 cm (given that potentiometer wire's resistance is 8Ω). Balance points are obtained with switch S opened and then closed, they are found at 50 cm and 25 cm respectively. Then, the unknown battery from circuit -1 is used in circuit -2 in order to deliver maximum possible power, to resistor R. The value of this maximum power will be x watt, find the integer value x.



circuit -1



circuit -2

18. Two balls of charge q_1 and q_2 initiallly have a velocity of the same magnitude and direction. A unifrom electric field has been applied for a certain time due to which the direction of the velocity of the first ball changes by 60^0, and the velocity magnitude is reduced by half. In the same time, the direction of the velocity of the second ball changes by 90^0. If the velocity of the second ball change such that the magnitude of the charge-to-mass ratio for the second ball is $(a/b)k_1$, where k_1 is charge-to-mass for the first ball. Then a + b is _____.
(The electrosatic interaction between the balls should be neglected.)

Space for Rough work

Chemistry

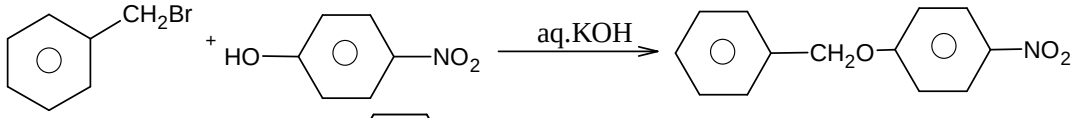
PART – II

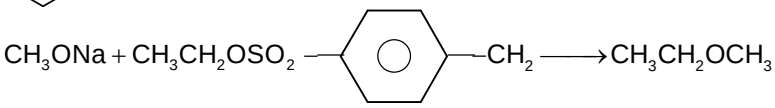
SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which of the following methods are useful for the synthesis of ether?

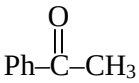
(A) $(CH_3)_3CBr + CH_3CH_2ONa \longrightarrow (CH_3)_3COCH_2CH_3$

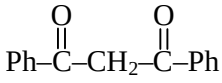
(B) 

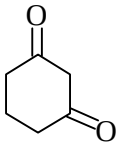
(C) 

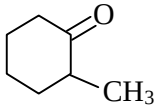
(D) $CH_3 - CH_2OH + CH_2N_2 \xrightarrow{BF_3} CH_3CH_2OCH_3 + N_2$

20. The compound which gives iodoform test is/are

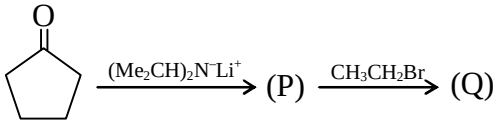
(A) 

(B) 

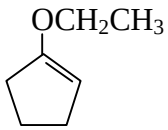
(C) 

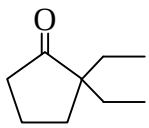
(D) 

21. The product(s) (Q) of the given reaction, in significant amounts

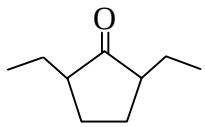


would be

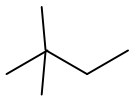
(A) 

(B) 

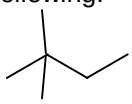
(C) $CH_2 = CH_2$

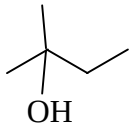
(D) 

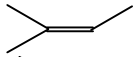
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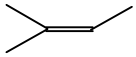
22. (X) $\xleftarrow[\text{reaction I}]{OH^-}$  $\xrightarrow[\text{reaction II}]{H_2O}$ (Y)

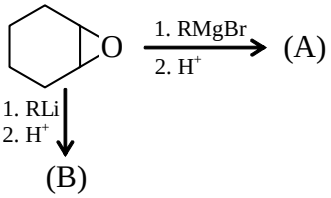
Choose the correct statement from the following:

(A) Minor product of reaction I would be  adopting S_N2 mechanism.

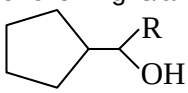
(B) Predominant product of reaction II would be  adopting S_N1 mechanism.

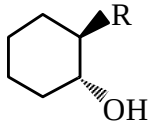
(C) Predominant product of reaction I would be  adopting E_1 mechanism.

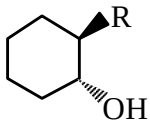
(D) Predominant product of reaction II would be  adopting E_1 mechanism.

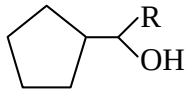
23. 

Which of the following is/are correct about structures of compound (A) and (B)?

(A) (A): 

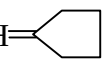
(B) (B): 

(C) (A): 

(D) (B): 

24. Which of the following reactions will produce acetaldehyde only (or as one of the product)?

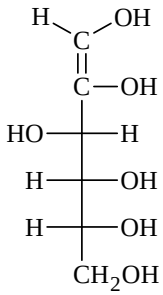
(A) $CH_2=CH-CH_2-CH=CH_2 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} ?$

(B) CH_3-CH  $\xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} ?$

(C) $CH \equiv CH + H_2O \xrightarrow[HgSO_4]{H_2SO_4} ?$

(D) $CH_3COCl + H_2 \xrightarrow[\text{quinoline}]{Pd\text{-}BaSO_4} ?$

Space for Rough work

25. 

The Fischer projection formula shown above is the enolic form of

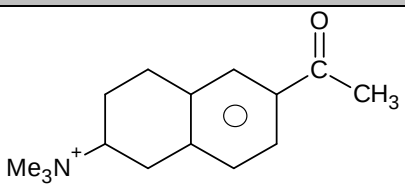
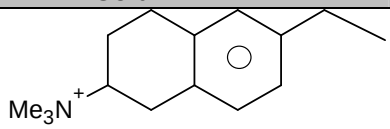
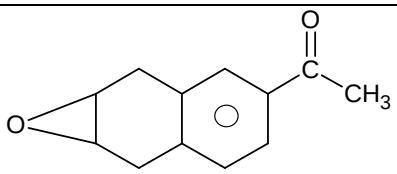
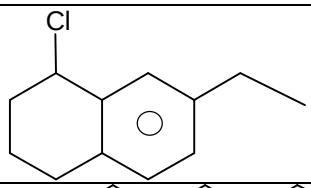
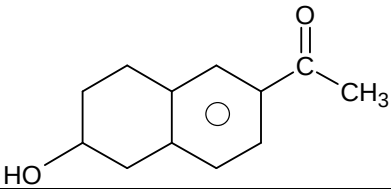
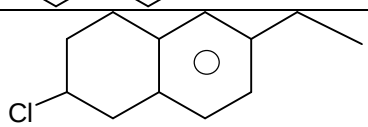
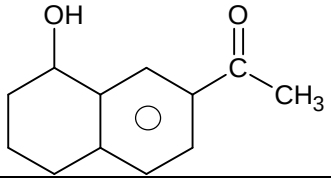
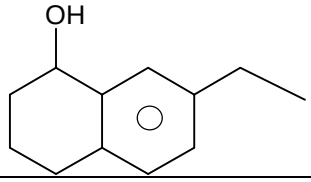
(A) D-fructose (B) D-mannose

(C) D-glucose (D) None of these

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions Q. 26, Q. 27 and Q. 28 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)		(i)		(P)	NH_2 / NH_2 / OH^-
(II)		(ii)		(Q)	$Zn(Hg)$, HCl
(III)		(iii)		(R)	$LiAlH_4$
(IV)		(iv)		(S)	None

26. Correct combination for (I)

(A) (I), (i), (Q) (B) (I) (i) (P)

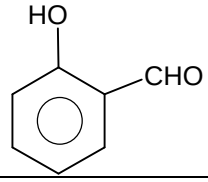
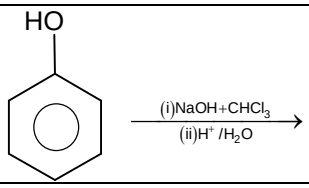
(C) (I), (i) (R) (D) (I) (i) (S)

Space for Rough work

27. Correct combination for (III)
(A) (III) (iii) (Q) (B) (III) (iv) (Q)
(C) (III), (iii) (P) (D) (III) (iii) (R)

28. Correct combination for (IV)
(A) (IV) (iii) (R) (B) (IV) (ii) (P)
(C) (IV), (iii) (Q) (D) (IV) (iv) (P)

Answer questions Q. 29, Q. 30 and Q. 31 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	$CH_3CH=CH_2 \xrightarrow[(ii)NaBH_4]{(i)(CH_3COO)_2Hg,H_2O}$	(i)		(P)	Williamson ether synthesis
(II)	$CH_3CH=CH_2 \xrightarrow[(ii)H_2O_2/OH]{(i)BH_3,THF}$	(ii)	$CH_3-CH_2-OCH_3$	(Q)	Reimer Tiemann Reaction
(III)	$CH_3CH_2ONa \xrightarrow{CH_3-Cl}$	(iii)	$CH_3-\underset{OH}{CH}-CH_3$	(R)	Oxymercuration demercuration
(IV)		(iv)	$CH_3-CH_2-CH_2OH$	(S)	Hydroboration oxidation

29. The only correct combination is
(A) (I) (iii) (S) (B) (II) (iv) (S)
(C) (III), (ii) (Q) (D) (IV) (i) (P)

30. Which one is only incorrect combination is
(A) (I) (iii) (R) (B) (II) (iv) (S)
(C) (III), (ii) (Q) (D) (IV) (i) (Q)

31. Which of the following is correct for Reimer Tiemann Reaction
(A) (I) (i) (Q) (B) (II) (iv) (Q)
(C) (IV), (i) (P) (D) (IV) (i) (Q)

Space for Rough work

SECTION – B
(Single digit integer type)

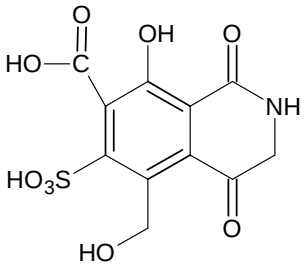
This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. It required 0.7 gm of hydrocarbon (A) to react completely with 2.0 gm of Br_2. On treatment of (A) with HBr it gave monobromoalkane (B). The same compound (B) was obtained when (A) was treated with HBr in presence of peroxide. Calculate the molecular weight of (B). Give your answer by dividing 68.5.

33. 0.535 g mixture of ethanol and acetaldehyde when heated with Fehling solution gave 1.2 g of a red precipitate. What is the percentage of acetaldehyde in the mixture? Give your answer by multiplying 100 and then divide by 6916.

34. In an amino acid the carboxylic acid group has $K_a = 10^{-4}$ and amino group has $K_a = 10^{-6}$ the isoelectric point of that amino acid is

35. How many moles of NaOH would be required for complete neutralization of following compound (1 mole)?



36. How many carbon atom in the final product

$$CH_3CH_2I \xrightarrow{NaCN} A \xrightarrow[\text{partial hydrolysis}]{OH^-} B \xrightarrow{NaOH + Br_2} C$$

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If $f(x) = \text{maximum}\{4, 1+x^2, x^2-1\}\ \forall x \in R$. Then the points where f(x) is non-differentiable are

(A) $\sqrt{3}$ (B) $-\sqrt{3}$

(C) Two irrational points (D) None

38. If $f''(x) > 0 \quad \forall x \in R, \quad f'(3) = 0$ and $g(x) = f(\tan^2 x - 2\tan x + 4), 0 < x < \frac{\pi}{2}$, then g(x) is increasing in

(A) $\left(0, \frac{\pi}{4}\right)$ (B) $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

(C) $\left(0, \frac{\pi}{3}\right)$ (D) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

39. A function f : R → R satisfies the equation $f(x+y) = f(x).f(y) \ \forall x, y \in R$ and $f(x) \neq 0$ for any x ∈ R. Let the function be differentiable at x = 0 and $f'(0) = 2$, then

(A) $f'(x) = 2f(x) \ \forall x \in R$ (B) f (x) is continuous everywhere

(C) $f(x) = e^{2x}$ (D) $f\left(\frac{1}{2}\right)$ is an irrational number

Space for Rough work

40. If P is a point on a circle of radius a and centre O (origin), the distance of chord AB from P lying along OP is x, then $\angle AOB \le \pi$ written as a function of x, has which of the following property

(A) $2\cos^{-1}\left(\frac{1}{a}|x|\right);0 < x < 2a$

(B) $2\sin^{-1}\left(\frac{1}{a}|x-a|\right);0 < x < 2a$

(C) $2\cos^{-1}\left(\frac{1}{a}|x-a|\right);0 < x < 2a$

(D) None of these

41. Let $f:[1,\infty) \to R$ is differentiable and satisfies $f'(x) = \frac{1}{\left(x^2 + f^2(x)\right)}$ and f(1) = 1, then

(A) $\lim_{x\to\infty} f(x) < 1 + \frac{\pi}{4}$

(B) $f'(x) < \frac{1}{1+x^2}$

(C) $\lim_{x\to\infty} f(x) < 1 + \frac{\pi}{3}$

(D) f (x) is monotonically increasing

42. The function $f(x) = \int_0^x \sqrt{1-t^4}\,dt$ is such that

(A) It is defined on the interval [-1, 1]

(B) It is an increasing function

(C) It is an odd function

(D) The point (0, 0) is the point of inflexion

43. If $I = \int \sec^2 x.\operatorname{cosec}^4 x\,dx = K\cot^3 x + L\tan x + M\cot x + C$, then

(A) $K = \frac{-1}{3}$

(B) L = 2

(C) M = -2

(D) $3K + L - M = 2$

Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions Q. 44, Q. 45 and Q. 46 by appropriately matching the information given in the three columns of the following table.

Column – 1 Function		Column – 2 Domain		Column – 3 Classification of function	
(I)	$f(x)=\sin(\sin x+\cos x)$	(i)	$(-\infty,\infty)$	(P)	Even
(II)	$f(x)=\dfrac{\sqrt{x+4}-3}{x-5}$	(ii)	$(10,\infty)$	(Q)	Odd
(III)	$f(x)=\dfrac{e^x-1}{e^x+1}$	(iii)	$(-\infty,-10)$	(R)	One-One
(IV)	$f(x)=\ell n(x^2+4x+5)$	(iv)	$(-10,20)$	(S)	Many-One

44. Which of the following combination is correct?
(A) (IV) (i) (S) (B) (I) (ii) (R)
(C) (II) (iii) (P) (D) (I) (iv) (R)

45. Which of the following combination is correct?
(A) (II) (iii) (S) (B) (II) (i) (Q)
(C) (III) (i) (R) (D) (II) (iii) (R)

46. Which of the following combination is NOT correct?
(A) (IV) (i) (S) (B) (I) (i) (S)
(C) (II) (ii) (R) (D) (III) (ii) (Q)

Space for Rough work

Answer questions Q. 47, Q. 48 and Q. 49 by appropriately matching the information given in the three columns of the following table.

Column 1 contains the function f(x)
Column 2 tells the continuity and differentiability of function at x = 0
Column 3 tells the limit of function at infinity

Column – 1		Column – 2		Column – 3	
(I)	$f(x) = x\sin\frac{1}{x},\ x \neq 0$	(i)	Continuous but not differentiable at x = 0	(P)	∞
(II)	$f(x) = \frac{x.e^{\frac{1}{x}}}{1+e^{\frac{1}{x}}},\ x \neq 0$	(ii)	Left continuous at x = 0	(Q)	0
(III)	$f(x) = \frac{1}{1-e^{\frac{-1}{x}}},\ x \neq 0$	(iii)	Continuous and differentiable at x = 0	(R)	1
(IV)	$f(x) = \frac{\frac{1}{e^x}-1}{e^{\frac{1}{x}}+1},\ x \neq 0$	(iv)	Right continuous at x = 0	(S)	-∞

47. Which of the following combination is correct?
(A) (I) (iii) (Q) (B) (I) (i) (R)
(C) (III) (ii) (Q) (D) (IV) (ii) (Q)

48. Which of the following combination is correct?
(A) (I) (ii) (Q) (B) (III) (ii) (P)
(C) (III) (iii) (S) (D) (II) (i) (Q)

49. Which of the following combination is correct?
(A) (IV) (iv) (Q) (B) (III) (i) (R)
(C) (IV) (ii) (Q) (D) (I) (iv) (R)

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. If $f(n+1)=\frac{1}{2}\left\{f(n)+\frac{9}{f(n)}\right\}$ where $n \in N$ and $f(n) > 0 \ \forall\, n \in N$ and $\lim_{n\to\infty} f(n)$ exist. Then the value of $\lim_{n\to\infty} f(n)$

51. Let F(x) be a non-negative continuous function defined on R such that $F(x)+F\left(x+\frac{1}{2}\right)=3$ and the value of $\int_0^{1500} F(x)dx$ is $\frac{9000}{\lambda}$. Then the value of λ is

52. Let f, g : R → R be the functions defined by $f(x)=x^2+1$ and $g(x)=2[x]-1$ where [x] is the greatest integer ≤ x. Find $(gofog)\left(\frac{2}{3}\right)$

53. If the graph of $f(x)=2x^3+ax^2+bx,\ \ a,b \in N$ cuts the x – axis at three real and distinct points, then the minimum value of a^2+b^2-4 is

54. If $f(x)=\begin{cases}\frac{[x]^2+\sin[x]}{[x]} & \text{for } [x]\neq 0\\ 0 & \text{for } [x]=0\end{cases}$, where $[x]$ denotes the greatest integer less than or equal to x, then $\left|\left[\lim_{x\to 0^-} f(x)\right]\right|$

Space for Rough work